Supplementary Tables for "Attribution-Constrained SMOTE: A Controlled Evaluation of Explanation-Guided Oversampling for Imbalanced Classification

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All tables are computed from the per-fold results released with the code (dataset-level means over 15 folds; $N = 10$ datasets). AC-SMOTE is compared with each baseline by a two-sided exact Wilcoxon signed-rank test. Holm and Benjamini–Hochberg (BH) adjustments are applied within a family of seven comparisons (one setting, one metric). Median diff. is the median dataset-level difference (AC-SMOTE minus baseline); r_{rb} is the matched-pairs rank-biserial correlation. Settings: LR = Logistic Regression on raw features; LR-std = standardized features (exploratory, seed 42).

S1. Equivalence tests: AC-SMOTE versus SMOTE

TOST: two one-sided Wilcoxon signed-rank tests against margins $\pm\delta$; the reported p is the larger of the two, Holm-adjusted across the four settings. CI: 95% percentile bootstrap interval of the median (20 000 resamples). Min. det. diff.: smallest difference detectable at 80% power by a paired t-test with the observed standard deviation of the differences, $N = 10$ (an approximation). Margins were chosen after the data were seen.

Metric	Setting	Median	CI low	CI high	Min. det.	$p_{0.01}$	$p_{0.02}$	SD diff.
G-mean	LR	+0.0007	-0.0013	+0.0051	0.0066	0.009	0.004	0.0066
G-mean	RF	+0.0010	-0.0109	+0.0075	0.0228	0.116	0.053	0.0230
G-mean	XGB	-0.0012	-0.0091	+0.0012	0.0128	0.084	0.004	0.0129
G-mean	LR-std	-0.0019	-0.0051	-0.0011	0.0034	0.008	0.004	0.0035
Balanced accuracy	LR	+0.0006	-0.0009	+0.0049	0.0058	0.009	0.004	0.0059
Balanced accuracy	RF	+0.0005	-0.0104	+0.0046	0.0098	0.024	0.004	0.0099
Balanced accuracy	XGB	-0.0010	-0.0072	+0.0008	0.0066	0.010	0.004	0.0066
Balanced accuracy	LR-std	-0.0018	-0.0051	-0.0010	0.0031	0.004	0.004	0.0031

S2. Pairwise comparisons: F1

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
LR	None	6/4/0	+0.0149	0.625	1.000	0.729	+0.20
	ROS	9/1/0	+0.0053	0.010	0.068	0.046	+0.89
	SMOTE	3/7/0	-0.0019	0.375	1.000	0.525	-0.35
	Borderline-SMOTE	7/3/0	+0.0054	0.846	1.000	0.846	+0.09
	ADASYN	8/2/0	+0.0102	0.020	0.098	0.046	+0.82
	SVM-SMOTE	1/9/0	-0.0154	0.014	0.082	0.046	-0.85
	SMOTE-Tomek	2/8/0	-0.0025	0.105	0.422	0.185	-0.60
RF	None	7/3/0	+0.0451	0.084	0.588	0.542	+0.64
	ROS	5/5/0	+0.0059	0.557	1.000	0.811	+0.24
	SMOTE	3/7/0	-0.0018	0.232	1.000	0.542	-0.45
	Borderline-SMOTE	5/5/0	-0.0015	0.922	1.000	0.922	-0.05
	ADASYN	6/4/0	+0.0025	0.160	0.961	0.542	+0.53
	SVM-SMOTE	3/7/0	-0.0041	0.625	1.000	0.811	-0.20
	SMOTE-Tomek	5/5/0	-0.0004	0.695	1.000	0.811	-0.16
XGB	None	7/3/0	+0.0191	0.084	0.588	0.588	+0.64
	ROS	6/4/0	+0.0033	0.695	1.000	0.770	+0.16
	SMOTE	3/7/0	-0.0012	0.375	1.000	0.770	-0.35
	Borderline-SMOTE	3/7/0	-0.0066	0.695	1.000	0.770	-0.16
	ADASYN	6/4/0	+0.0034	0.375	1.000	0.770	+0.35
	SVM-SMOTE	4/6/0	-0.0057	0.770	1.000	0.770	-0.13
	SMOTE-Tomek	5/5/0	+0.0008	0.770	1.000	0.770	+0.13
LR-std	None	5/5/0	+0.0032	0.922	1.000	0.922	+0.05

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
	ROS	4/6/0	-0.0006	0.492	1.000	0.689	-0.27
	SMOTE	0/10/0	-0.0037	0.002	**0.014**	0.007	-1.00
	Borderline-SMOTE	5/5/0	+0.0010	0.695	1.000	0.811	-0.16
	ADASYN	7/3/0	+0.0054	0.084	0.336	0.147	+0.64
	SVM-SMOTE	0/10/0	-0.0180	0.002	**0.014**	0.007	-1.00
	SMOTE-Tomek	1/9/0	-0.0042	0.004	**0.020**	0.009	-0.96

S3. Pairwise comparisons: AUC-PR

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
LR	None	4/6/0	-0.0074	0.557	1.000	0.649	-0.24
	ROS	8/2/0	+0.0020	0.232	0.930	0.407	+0.45
	SMOTE	4/6/0	-0.0014	0.375	1.000	0.525	-0.35
	Borderline-SMOTE	9/1/0	+0.0066	0.014	0.096	0.096	+0.85
	ADASYN	8/2/0	+0.0054	0.064	0.322	0.150	+0.67
	SVM-SMOTE	5/5/0	+0.0001	0.846	1.000	0.846	+0.09
	SMOTE-Tomek	3/7/0	-0.0041	0.027	0.164	0.096	-0.78
RF	None	4/6/0	-0.0066	0.375	1.000	0.525	-0.35
	ROS	3/7/0	-0.0030	0.322	1.000	0.525	-0.38
	SMOTE	5/5/0	+0.0007	0.695	1.000	0.811	-0.16
	Borderline-SMOTE	8/2/0	+0.0068	0.084	0.588	0.525	+0.64
	ADASYN	6/4/0	+0.0028	0.275	1.000	0.525	+0.42
	SVM-SMOTE	4/6/0	-0.0037	0.375	1.000	0.525	-0.35
	SMOTE-Tomek	5/5/0	+0.0007	1.000	1.000	1.000	+0.02
XGB	None	6/4/0	+0.0029	0.695	1.000	0.846	+0.16
	ROS	8/2/0	+0.0028	0.193	1.000	0.677	+0.49
	SMOTE	4/6/0	-0.0007	0.846	1.000	0.846	-0.09
	Borderline-SMOTE	7/3/0	+0.0041	0.322	1.000	0.752	+0.38
	ADASYN	9/1/0	+0.0041	0.004	**0.027**	0.027	+0.96
	SVM-SMOTE	4/6/0	-0.0094	0.492	1.000	0.846	-0.27
	SMOTE-Tomek	4/6/0	-0.0026	0.770	1.000	0.846	-0.13
LR-std	None	1/9/0	-0.0111	0.014	0.096	0.096	-0.85
	ROS	8/2/0	+0.0016	0.275	1.000	0.386	+0.42
	SMOTE	3/7/0	-0.0008	0.160	0.961	0.374	-0.53
	Borderline-SMOTE	7/3/0	+0.0040	0.492	1.000	0.574	+0.27
	ADASYN	6/4/0	+0.0027	0.275	1.000	0.386	+0.42
	SVM-SMOTE	6/4/0	+0.0002	0.695	1.000	0.695	-0.16
	SMOTE-Tomek	3/7/0	-0.0006	0.160	0.961	0.374	-0.53

S4. Pairwise comparisons: MCC

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
LR	None	6/4/0	+0.0061	0.557	1.000	0.649	+0.24
	ROS	9/1/0	+0.0055	0.020	0.137	0.068	+0.82
	SMOTE	3/7/0	-0.0018	0.770	1.000	0.770	-0.13
	Borderline-SMOTE	7/3/0	+0.0095	0.492	1.000	0.649	+0.27
	ADASYN	9/1/0	+0.0119	0.020	0.137	0.068	+0.82
	SVM-SMOTE	2/8/0	-0.0094	0.064	0.322	0.150	-0.67
	SMOTE-Tomek	2/8/0	-0.0020	0.275	1.000	0.482	-0.42
RF	None	6/4/0	+0.0221	0.275	1.000	0.643	+0.42
	ROS	5/5/0	-0.0005	1.000	1.000	1.000	+0.02
	SMOTE	2/8/0	-0.0015	0.193	1.000	0.643	-0.49
	Borderline-SMOTE	5/5/0	-0.0028	0.770	1.000	1.000	-0.13
	ADASYN	7/3/0	+0.0019	0.105	0.738	0.643	+0.60
	SVM-SMOTE	3/7/0	-0.0109	0.492	1.000	0.861	-0.27
	SMOTE-Tomek	5/5/0	-0.0008	0.922	1.000	1.000	-0.05
XGB	None	6/4/0	+0.0147	0.432	1.000	0.846	+0.31
	ROS	6/4/0	+0.0037	0.770	1.000	0.846	+0.13
	SMOTE	4/6/0	-0.0022	0.492	1.000	0.846	-0.27

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
	Borderline-SMOTE	3/7/0	-0.0071	0.846	1.000	0.846	-0.09
	ADASYN	7/3/0	+0.0044	0.193	1.000	0.846	+0.49
	SVM-SMOTE	5/5/0	-0.0046	0.846	1.000	0.846	-0.09
	SMOTE-Tomek	6/4/0	+0.0039	0.557	1.000	0.846	+0.24
LR-std	None	4/6/0	-0.0100	0.922	1.000	0.922	-0.05
	ROS	3/7/0	-0.0009	0.432	1.000	0.604	-0.31
	SMOTE	0/10/0	-0.0040	0.002	**0.014**	0.014	-1.00
	Borderline-SMOTE	6/4/0	+0.0060	0.557	1.000	0.649	+0.24
	ADASYN	7/3/0	+0.0066	0.131	0.523	0.229	+0.56
	SVM-SMOTE	1/9/0	-0.0188	0.020	0.098	0.046	-0.82
	SMOTE-Tomek	1/9/0	-0.0040	0.004	**0.023**	0.014	-0.96

S5. Pairwise comparisons: G-mean

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
LR	None	10/0/0	+0.2577	0.002	**0.014**	0.014	+1.00
	ROS	6/4/0	+0.0009	0.275	1.000	0.482	+0.42
	SMOTE	6/4/0	+0.0007	0.492	1.000	0.574	+0.27
	Borderline-SMOTE	9/1/0	+0.0043	0.037	0.223	0.130	+0.75
	ADASYN	8/2/0	+0.0030	0.105	0.527	0.246	+0.60
	SVM-SMOTE	6/4/0	+0.0024	0.492	1.000	0.574	+0.27
	SMOTE-Tomek	5/5/0	+0.0004	0.770	1.000	0.770	+0.13
RF	None	10/0/0	+0.1151	0.002	**0.014**	0.014	+1.00
	ROS	8/2/0	+0.0508	0.037	0.223	0.087	+0.75
	SMOTE	6/4/0	+0.0010	1.000	1.000	1.000	+0.02
	Borderline-SMOTE	7/3/0	+0.0118	0.037	0.223	0.087	+0.75
	ADASYN	4/6/0	-0.0010	0.846	1.000	0.987	-0.09
	SVM-SMOTE	4/6/0	-0.0063	0.695	1.000	0.973	-0.16
	SMOTE-Tomek	4/6/0	-0.0115	0.275	1.000	0.482	-0.42
XGB	None	10/0/0	+0.0501	0.002	**0.014**	0.014	+1.00
	ROS	7/3/0	+0.0163	0.084	0.504	0.294	+0.64
	SMOTE	4/6/0	-0.0012	0.375	1.000	0.656	-0.35
	Borderline-SMOTE	5/5/0	-0.0000	1.000	1.000	1.000	-0.02
	ADASYN	5/5/0	+0.0004	0.922	1.000	1.000	+0.05
	SVM-SMOTE	4/6/0	-0.0059	0.275	1.000	0.643	-0.42
	SMOTE-Tomek	4/6/0	-0.0030	0.492	1.000	0.689	-0.27
LR-std	None	10/0/0	+0.1819	0.002	**0.014**	0.007	+1.00
	ROS	2/8/0	-0.0009	0.232	0.697	0.325	-0.45
	SMOTE	1/9/0	-0.0019	0.006	**0.029**	0.014	-0.93
	Borderline-SMOTE	7/3/0	+0.0063	0.037	0.148	0.065	+0.75
	ADASYN	6/4/0	+0.0011	0.695	1.000	0.695	+0.16
	SVM-SMOTE	4/6/0	-0.0011	0.625	1.000	0.695	+0.20
	SMOTE-Tomek	0/10/0	-0.0022	0.002	**0.014**	0.007	-1.00

S6. Pairwise comparisons: Balanced accuracy

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
LR	None	10/0/0	+0.1816	0.002	**0.014**	0.014	+1.00
	ROS	7/3/0	+0.0013	0.160	0.641	0.280	+0.53
	SMOTE	6/4/0	+0.0006	0.432	1.000	0.604	+0.31
	Borderline-SMOTE	9/1/0	+0.0032	0.037	0.223	0.130	+0.75
	ADASYN	8/2/0	+0.0028	0.105	0.527	0.246	+0.60
	SVM-SMOTE	5/5/0	-0.0005	0.922	1.000	0.922	+0.05
	SMOTE-Tomek	5/5/0	+0.0004	0.695	1.000	0.811	+0.16
RF	None	10/0/0	+0.0572	0.002	**0.014**	0.014	+1.00
	ROS	8/2/0	+0.0291	0.027	0.137	0.064	+0.78
	SMOTE	5/5/0	+0.0005	0.922	1.000	1.000	-0.05
	Borderline-SMOTE	8/2/0	+0.0097	0.020	0.117	0.064	+0.82
	ADASYN	4/6/0	-0.0006	1.000	1.000	1.000	-0.02

Setting	Baseline	W/L/T	Median diff.	p (raw)	p (Holm)	p (BH)	r_{rb}
	SVM-SMOTE	3/7/0	-0.0051	0.625	1.000	0.875	-0.20
	SMOTE-Tomek	3/7/0	-0.0053	0.275	1.000	0.482	-0.42
XGB	None	9/1/0	+0.0386	0.004	**0.027**	0.027	+0.96
	ROS	7/3/0	+0.0131	0.084	0.504	0.294	+0.64
	SMOTE	3/7/0	-0.0010	0.232	1.000	0.407	-0.45
	Borderline-SMOTE	4/6/0	-0.0005	1.000	1.000	1.000	+0.02
	ADASYN	5/5/0	+0.0005	0.922	1.000	1.000	-0.05
	SVM-SMOTE	4/6/0	-0.0027	0.432	1.000	0.604	-0.31
	SMOTE-Tomek	4/6/0	-0.0026	0.232	1.000	0.407	-0.45
LR-std	None	10/0/0	+0.1428	0.002	**0.014**	0.007	+1.00
	ROS	2/8/0	-0.0009	0.232	0.697	0.325	-0.45
	SMOTE	1/9/0	-0.0018	0.006	**0.029**	0.014	-0.93
	Borderline-SMOTE	8/2/0	+0.0049	0.037	0.148	0.065	+0.75
	ADASYN	5/5/0	-0.0003	0.770	1.000	0.770	+0.13
	SVM-SMOTE	4/6/0	-0.0012	0.695	1.000	0.770	+0.16
	SMOTE-Tomek	0/10/0	-0.0024	0.002	**0.014**	0.007	-1.00

Bold: Holm-adjusted $p < 0.05$ within its family of seven.